11.11 Classwork: Regression and Normal Distribution (calculator) — Markscheme

1. [Maximum mark: 7]

A sports scientist is studying the relationship between the age of an athlete, a years, and their reaction time, r milliseconds. The following table shows the data collected from eight athletes.

a	18	21	25	28	33	36	40	45
r	155	159	176	178	198	199	215	226

(a) Find the regression line of r on a in the form $r = ma + c$, where m and c are constants. Give m and c correct to three significant figures. [3]

GDC: Enter data into lists, run linear regression.

$m = 2.70$ **(A1)**

$c = 105$ **(A1)**

$r = 2.70a + 105$ **(A1)**

*Award **(A1)** for each of m and c correct to 3 s.f., and **(A1)** for writing the equation in the required form. Accept $m = 2.70$ (3 s.f.) and $c = 105$ (3 s.f.).*

(b) Write down the value of Pearson's product-moment correlation coefficient, r. [1]

$r = 0.994$ **(A1)**

Accept 0.99 *(2 s.f.) or* 0.9937 *(4 s.f.).*

(c) Use the regression line to estimate the reaction time of an athlete who is 30 years old. [2]

$r = 2.70(30) + 105$ **(M1)**

$= 186$ ms **(A1)**

*Award **(M1)** for substituting $a = 30$ into their regression equation. Accept* 186 *to* 186.2 *depending on rounding of m and c.*

(d) State whether this estimate is an example of interpolation or extrapolation. Give a reason for your answer. [1]

Interpolation, because 30 is within the range of the data ($18 \le a \le 45$). **(R1)**

*Award **(R1)** for "interpolation" with a correct reason referencing the data range. Do not award if the reason is missing or incorrect.*

2. [Maximum mark: 8]

The time, T minutes, that customers wait in line at a coffee shop is normally distributed with a mean of 4.5 minutes and a standard deviation of 1.2 minutes.

(a) Find $\mathrm{P}(T > 6)$. [2]

GDC: normalcdf with lower $= 6$*, upper* $= \infty$*,* $\mu = 4.5$*,* $\sigma = 1.2$.

$\mathrm{P}(T > 6) = 1 - \mathrm{P}(T \leq 6)$ **(M1)**

$= 0.106$ **(A1)**

*Award **(M1)** for a correct probability statement or GDC setup. Accept* 0.1056 *or* 0.106.

(b) Find $\mathrm{P}(3 < T < 6)$. [2]

GDC: normalcdf with lower $= 3$*, upper* $= 6$*,* $\mu = 4.5$*,* $\sigma = 1.2$.

$\mathrm{P}(3 < T < 6) = \mathrm{P}(T < 6) - \mathrm{P}(T < 3)$ **(M1)**

$= 0.789$ **(A1)**

*Award **(M1)** for a correct approach (difference of two probabilities or direct GDC entry). Accept* 0.7887 *or* 0.789.

(c) On a particular morning, 10 customers each wait in line independently. Find the probability that exactly 2 of them wait more than 6 minutes. [2]

Let $X \sim \mathrm{B}(10,\ 0.1056\ldots)$ **(M1)**

GDC: binompdf(10, 0.1056, 2)

$\mathrm{P}(X = 2) = 0.206$ **(A1)**

*Award **(M1)** for recognizing a binomial distribution with* $n = 10$ *and* p *from part (a). Award **(A1)** for a correct answer. Follow through from their answer to part (a). Accept* 0.206 *or* 0.2056.

(d) A customer is selected at random. Given that they waited more than 3 minutes, find the probability that they waited more than 6 minutes. [2]

$$\mathrm{P}(T > 6 \mid T > 3) = \frac{\mathrm{P}(T > 6)}{\mathrm{P}(T > 3)}$$ **(M1)**

$\mathrm{P}(T > 3) = 0.894$

$$= \frac{0.1056\ldots}{0.8944\ldots} = 0.118$$ **(A1)**

*Award **(M1)** for correct use of the conditional probability formula. Award **(A1)** for a correct answer. Accept* 0.118 *or* 0.1181.